

Citizen Grievance & Sentiment Analysis System

SUBMITTED BY UVESH SHAIKH

Week 1 Report: Data Collection, Cleaning and Exploratory Data Analysis

1. Introduction

The objective of this project is to develop an AI-based system for analyzing citizen complaints in the government and public sector. The system aims to automatically categorize complaints and analyze their sentiment to improve grievance redressal.

2. Data Collection and Understanding

The dataset contains structured information about citizen complaints collected from public service records, including details such as complaint type, description, location, and status. These attributes help in understanding the nature of public issues.

[1]:

```
import pandas as pd

df = pd.read_csv("C:\\Users\\UVESH SHAIKH\\OneDrive\\Pictures\\Desktop\\DS&ML\\Project-1\\clipdataset.csv")
print(df.columns)
```

C:\\Temp\\ipykernel_25940\\3557705156.py:3: DtypeWarning: Columns (31) have mixed types. Specify dtype option on import or set low_memory=False.
df = pd.read_csv("C:\\Users\\UVESH SHAIKH\\OneDrive\\Pictures\\Desktop\\DS&ML\\Project-1\\clipdataset.csv")
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
'Intersection Street 1', 'Intersection Street 2', 'Address Type',
'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
'Resolution Description', 'Resolution Action Updated Date',
'Community Board', 'BBL', 'Borough', 'X Coordinate (State Plane)',
'Y Coordinate (State Plane)', 'Open Data Channel Type',
'Park Facility Name', 'Park Borough', 'Vehicle Type',
'Taxi Company Borough', 'Taxi Pick Up Location', 'Bridge Highway Name',
'Bridge Highway Direction', 'Road Ramp', 'Bridge Highway Segment',
'Latitude', 'Longitude', 'Location', 'Zip Codes', 'Community Districts',
'Borough Boundaries', 'City Council Districts', 'Police Precincts'],
dtype='object')

[2]:

```
df.head()
```

[2]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Location Type	Incident Zip	Incident Address	...	Road Ramp	Bridge Highway Segment	Latitude
0	45052849	2019-12-01T01:42:46.000	NaN	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Store/Commercial	10002.0	151 RIVINGTON STREET	...	NaN	NaN	40.71924
1	45051451	2019-12-01T01:24:47.000	NaN	NYPD	New York City Police Department	Noise - Residential	Banging/Pounding	NaN	10452.0	1675 GRAND CONCOURSE	...	NaN	NaN	40.84446
2	45049327	2019-12-01T01:04:16.000	NaN	NYPD	New York City Police Department	Illegal Parking	Blocked Hydrant	Street/Sidewalk	11374.0	65-41 SAUNDERS STREET	...	NaN	NaN	40.72756
3	45053522	2019-12-01T00:48:01.000	NaN	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	NaN	11435.0	90 AVENUE	...	NaN	NaN	40.70127
4	45050836	2019-11-30T23:22:48.000	2019-12-01T01:27:32.000	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Street/Sidewalk	10307.0	CRAIG AVENUE	...	NaN	NaN	40.51175

5 rows x 46 columns

2.1 Data Information

The dataset consists of multiple columns representing different aspects of complaints. Important features include:

- Complaint Type: Type/category of complaint
- Descriptor: Detailed description of the issue
- City: Location of the complaint
- Status: Current state of complaint resolution
- Created Date and Closed Date: Timeline information

The dataset contains both categorical and numerical data. Some columns have missing values, especially in descriptive fields, which were handled during preprocessing.

Source Code:

```
df.info()
df.shape
df.isnull().sum()
```

3. Data Preprocessing (Text Cleaning)

Since the dataset does not contain a single text column, a new column named “**text**” was created by combining the “Complaint Type” and “Descriptor” columns.

Source Code:

```
df['text'] = df['Complaint Type'].fillna("") + " " + df['Descriptor'].fillna("")
```

Text cleaning was performed to make the data suitable for analysis. The following steps were applied:

- Converted text to lowercase
- Removed special characters and numbers
- Removed stopwords such as “is”, “the”, and “and”

```
[7]: import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

nltk.download('stopwords')
nltk.download('wordnet')

lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))

[nltk_data] Downloading package stopwords to C:\Users\UVESH
[nltk_data]   SHAIKH\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to C:\Users\UVESH
[nltk_data]   SHAIKH\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!

[8]: df['text'] = df['Complaint Type'] + " " + df['Descriptor']
df['text'] = df['Complaint Type'].fillna('') + " " + df['Descriptor'].fillna('')

df['clean_text'] = df['text'].str.lower()

df['clean_text'] = df['clean_text'].apply(lambda x: re.sub(r'^a-zA-Z ', '', x))

df['clean_text'] = df['clean_text'].apply(
    lambda x: " ".join([word for word in x.split() if word not in stop_words])
)
df[['text', 'clean_text']].head()
```

```
import re
from nltk.corpus import stopwords
```

```
stop_words = set(stopwords.words('english'))

df['clean_text'] = df['text'].str.lower()

df['clean_text'] = df['clean_text'].apply(lambda x: re.sub(r'^a-zA-Z ', '', x))

df['clean_text'] = df['clean_text'].apply(
    lambda x: " ".join([word for word in x.split() if word not in stop_words])
)

df[['text', 'clean_text']].head()
```

4. Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns and trends in the complaint data.

4.1 WordCloud Visualization

A WordCloud was generated to visualize the most frequent words in the dataset. Words appearing larger indicate higher frequency.

```
from wordcloud import WordCloud

text_data = " ".join(df['clean_text'])

wordcloud = WordCloud(width=800, height=400, background_color='white').generate(text_data)

plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Common Complaints WordCloud")
plt.show()
```



The WordCloud shows frequently occurring terms such as “noise”, “water”, and “street”, indicating major urban issues. The bar graph highlights the most common complaint categories, helping identify priority areas for government action.

4.2 Complaint Type Distribution

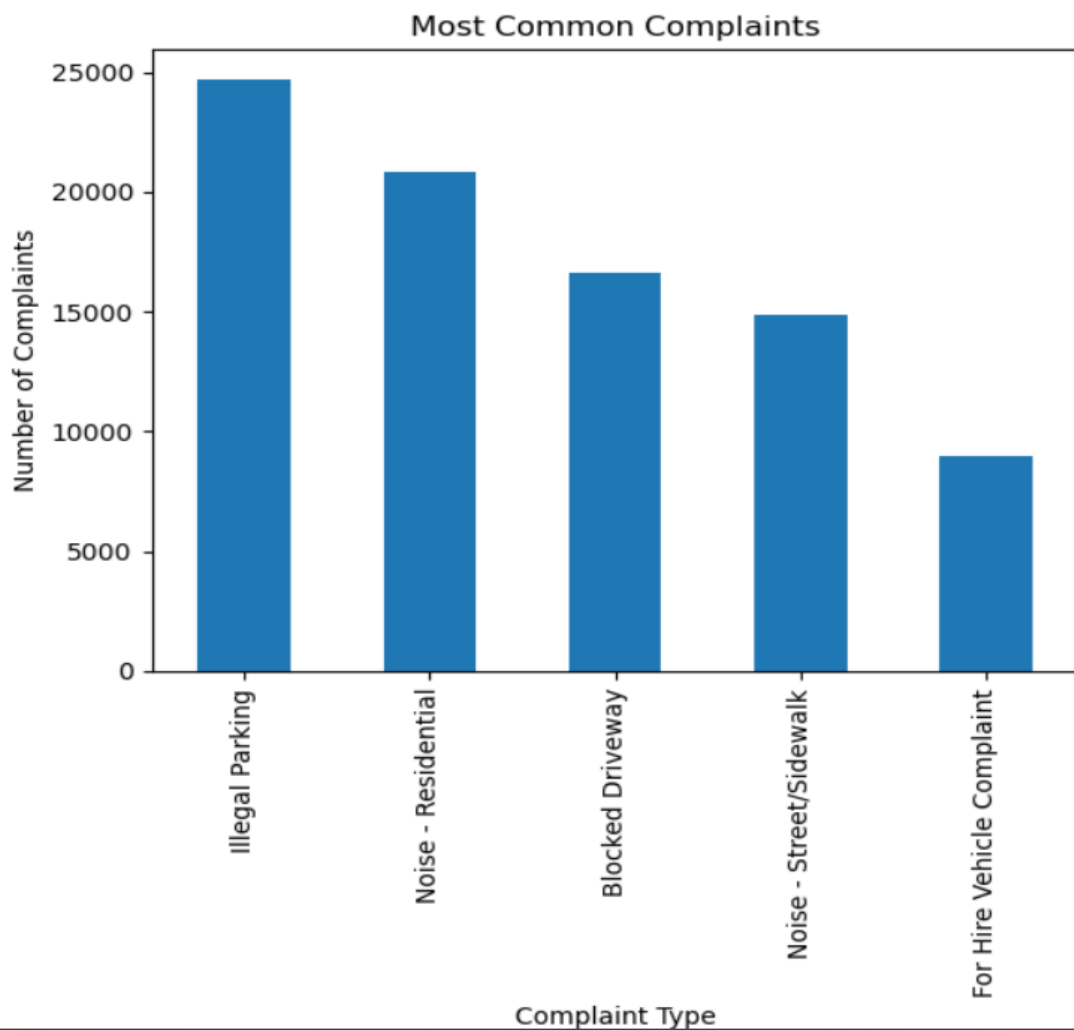
The bar graph clearly shows the most frequent complaint types, helping identify major areas that require government attention.

```
: import matplotlib.pyplot as plt

df['Complaint Type'].value_counts().head(5).plot(kind='bar')

plt.title("Most Common Complaints")
plt.xlabel("Complaint Type")
plt.ylabel("Number of Complaints")

plt.show()
```



5. Key Insights

- Most complaints are related to urban infrastructure and public issues
- Common issues include noise, water supply, and street-related problems
- Text preprocessing improves the quality of data for further analysis
- Visualization techniques help in identifying major complaint trends

6. Conclusion

In Week 1, the dataset was successfully analyzed and prepared for further processing. Text preprocessing techniques were applied to clean the data, and exploratory data analysis was performed to understand complaint patterns. Visualizations such as WordCloud and bar charts provided meaningful insights into common public issues. This work forms the foundation for building machine learning models for complaint classification and sentiment analysis in the upcoming weeks.